

Supplementary Information

LongPhase 2: uncertainty-aware co-phasing of small variants, structural variants and DNA methylation from long reads

Jyun-Hong Lin, [\[co-authors\]](#), Yao-Ting Huang

Contents

Supplementary Notes	2
Supplementary Tables	3
Supplementary Figures	9

Supplementary Notes

Supplementary Note 1. Reading the phasing graph export

With `-dot`, `longphase phase` writes one file per chromosome, `<prefix>.<chrom>.dot`, containing one line per voting pair, `P.a -> Q.b [label=w]`, where `P` and `Q` are 1-based positions, `a`, `b` $\in \{1, 2\}$ index the reference (1) and alternate (2) allele, and `w` is the weighted haplotype vote accumulated at `P` for that allele at the time the vote was cast. The companion VCF carries, for every phased variant, the fields `INFO/H1`, `INFO/H2` (weighted votes for haplotype 1 and 2) and `INFO/PE` (their normalized Shannon entropy). The `gnn` sub-command reads both files; users who wish to inspect why a variant was unphased can therefore reconstruct its local window without re-running the phaser.

Supplementary Note 2. Reproducing the demonstration

The repository ships a 500-kb demonstration (`chr1:121.5–122.0 Mb`; `demo/demo.sh`) that runs `modcall`, SNV-only and four-class `phase`, `gnn` correction of both outputs and `compare` against a truth VCF, and prints switch errors, phased SNV counts and Hamming distance for the four configurations. Total runtime is under two minutes on four threads.

Supplementary Tables

Supplementary Table 1: Node features of the graph neural network. Each allele node of a window carries 31 features; # is the index in the feature vector. h_1, h_2 : weighted haplotype votes (taken from the DOT edge labels when available, otherwise from INFO/H1, INFO/H2); PE: phasing entropy; PS: phase set; “centre” is the trigger variant of the window. Genomic-context features are computed from the reference when -r is given and are zero otherwise.

#	Feature	Definition
<i>Phasing evidence (16)</i>		
1	PE	normalized Shannon entropy of (h_1, h_2) , 0–1 bit
2	HP	haplotype of this allele node (genotype index)
3	gt_ref	genotype index of the reference allele (0)
4	gt_alt	genotype index of the alternate allele (1)
5	is_center	1 if the variant is the window trigger
6	is_same_ps	1 if the variant shares the trigger’s PS
7	allele	0 for the reference-allele node, 1 for the alternate-allele node
8	rel_offset	$(\text{pos} - \text{pos}_c) / \max \text{pos} - \text{pos}_c $ over the window, signed
9	H1	h_1
10	H2	h_2
11	total_votes	$h_1 + h_2$
12	vote_ratio	$\max(h_1, h_2) / (h_1 + h_2)$
13	log_total	$\log_2(1 + h_1 + h_2)$
14	log_h1	$\log_2(1 + h_1)$
15	log_h2	$\log_2(1 + h_2)$
25	rel_vote_depth	$\log_2((h_1 + h_2) / \overline{(h_1 + h_2)}_{\pm 10 \text{ variants}})$
<i>Genomic context (6)</i>		
16	hp_len	homopolymer run length at the variant (capped at 20)
17	gc_content	GC fraction in ± 50 bp
18	str_context	longest tandem-repeat unit count (unit 2–4 bp) in ± 15 bp, /10, capped at 1
19	seq_entropy	Shannon entropy of the ± 15 bp sequence (bits)
20	neigh_dens	$(\text{number of variants within } \pm 500 \text{ bp} - 1) / 20$
21	dist_near_indel	$\log_{10}(1 + \text{distance to nearest indel})$, capped at $\log_{10}(10001)$
<i>Graph structure (5)</i>		
22	out_degree	number of distinct target positions of outgoing DOT edges, /10, capped at 1
23	ps_block_size	$\log(1 + \text{variants in the PS}) / \log(101)$
24	log_center_dist	$\log_{10}(1 + \text{pos} - \text{pos}_c)$
26	out_weight_std	s.d. of outgoing DOT edge weights, /100, capped at 1
27	is_bridge	1 if removing the variant disconnects the window graph
<i>Variant class (4, one-hot)</i>		
28–31	is_SNV, is_indel, is_SV, is_5mC	variant class indicators

Supplementary Table 2: Edge features of the graph neural network. Edges are taken from the DOT export, restricted to node pairs inside the window, symmetrized, and complemented by one self-loop per node whose features are the mean over the node’s incident edges. w is the DOT edge label (accumulated vote weight).

#	Feature	Definition
1	weight	$1/(1 + e^{-w})$
2	log_dist	$\log_{10}(1 + \text{pos}_{\text{dst}} - \text{pos}_{\text{src}})$
3	log_weight	$\log(1 + w)$
4	is_same_hp	1 if the two allele nodes have the same allele index
5	is_inter_ps	1 if the two variants belong to different phase sets
6	weight_ratio	$w / (\text{sum of outgoing edge weights of the source allele node})$

Supplementary Table 3: Architecture and inference hyperparameters of the graph neural network.

Component	Setting
Input normalization	BatchNorm over 31 node features (frozen statistics at inference)
Input projection	Linear $31 \rightarrow 128$, GELU
Layers	4 GPS layers, hidden size 128
Local branch	GATv2, 4 heads $\times$ 32, LeakyReLU slope 0.2, per-layer edge encoder $6 \rightarrow 128$
Global branch	multi-head self-attention, 4 heads, Q/K/V/O projections $128 \rightarrow 128$
Fusion	$x \leftarrow x + \text{local} + \text{global}$; LayerNorm ($\varepsilon = 10^{-5}$)
Feed-forward	$128 \rightarrow 256 \rightarrow 128$, GELU, residual, LayerNorm
Classifier	$\text{concat}[h_i, \sum_j w_{ij}, \max_j w_{ij}] (130) \rightarrow 128$ (GELU) $\rightarrow 2$, softmax
Parameters	687,358 (float32, 2.75 MB, embedded in the binary)
Window	trigger $\text{PE} \geq 0.80$; ± 20 variants; ≤ 256 nodes
Decision	unphase if mean $P(\text{error}) \geq 0.30$ over the two allele nodes and all covering windows
Post-processing	split disconnected phase sets (largest component keeps PS); bridge protection off
Training	[data, labels, loss, optimizer, epochs]

Supplementary Table 4: Parameters of longphase phase and their defaults.

Parameter	Default	Meaning
-ont / -pb	–	sequencing platform (enables nanopore-specific homopolymer filter)
-indels	off	phase small indels
-indelQuality	0	exclude indels with QUAL below this value
-sv-file, -mod-file	–	SV VCF (Sniffles2/cuteSV with read names); 5mC VCF from modcall
-q, -mappingQuality	1	minimum mapping quality of alignments
-x, -mismatchRate	3	mark reads as unreliable above this mismatch rate
-p, -baseQuality	12	base quality below which an observation is down-weighted
-e, -edgeWeight	0.1	weight of a down-weighted observation (otherwise 1)
-a, -connectAdjacent	35	number of downstream variants each variant is linked to
-d, -distance	300,000	maximum distance (bp) between linked variants
-l, -edgeThreshold	0.7	pair casts no vote if $\min(P, Q) / \max(P, Q)$ exceeds this (0.3 for SNV–5mC)
-L, -overlapThreshold	0.2	overlap fraction above which the shorter of two alignments of a read is dropped
-w, -sv-window	20	CIGAR operations scanned around an SV start
-h, -sv-threshold	0.10	relative length tolerance for a CIGAR indel to support an SV
-m, -readConfidence	0.65	minimum majority fraction to tag a read in read-based correction
-n, -snpConfidence	0.75	minimum confidence to keep a variant’s phase after read-based correction
-dot	off	export the phasing graph (required by gnn)

Supplementary Table 5: Parameters of `longphase modcall` and their defaults. Values are those set in the source; the on-line help text lists 0.7 for `-heterRatio` and 6 for `-connectAdjacent` ([\[reconcile\]](#)).

Parameter	Default	Meaning
<code>-m, -modThreshold</code>	0.8	ML probability at or above which a base is methylated
<code>-u, -unModThreshold</code>	0.2	ML probability at or below which a base is unmethylated
<code>-e, -heterRatio</code>	0.6	minimum $\min(m, u) / \max(m, u)$ for a heterozygous site
<code>-i, -noiseRatio</code>	0.2	maximum fraction of reads with intermediate probability
<code>-a, -connectAdjacent</code>	20	neighbouring variants searched for SNV co-segregation
<code>-c, -connectConfidence</code>	0.9	minimum majority ratio of co-segregating reads
<code>-all</code>	off	also output homozygous modification sites

Supplementary Table 6: Parameters of `longphase gnn` and `longphase haplotag`.

Parameter	Default	Meaning
<i>gnn</i>		
<code>-s, -snp-file</code>	–	phased SNV/indel VCF from <code>phase</code>
<code>-sv-file, -mod-file</code>	–	phased SV and 5mC VCFs (corrected copies are written)
<code>-dot-prefix</code>	from <code>-s</code>	prefix of the DOT files
<code>-r, -reference</code>	–	reference FASTA (enables genomic-context features)
<code>-pe-threshold</code>	0.80	phasing entropy that opens a window
<code>-window</code>	20	variants on each side of the trigger
<code>-B, -break-threshold</code>	0.30	error probability at or above which a variant is unphased
<code>-respect-bridge</code>	off	never unphase bridge vertices
<code>-no-split-blocks</code>	off	keep the original PS even if the block becomes disconnected
<i>haplotag</i>		
<code>-q, -qualityThreshold</code>	1	minimum mapping quality of tagged alignments
<code>-p, -percentageThreshold</code>	0.6	minimum fraction of a read's informative alleles on the majority haplotype
<code>-tagSupplementary</code>	off	also tag supplementary alignments
<code>-sv-file, -mod-file</code>	–	use phased SVs and 5mC sites as additional tagging evidence
<code>-cram</code>	off	write CRAM instead of BAM

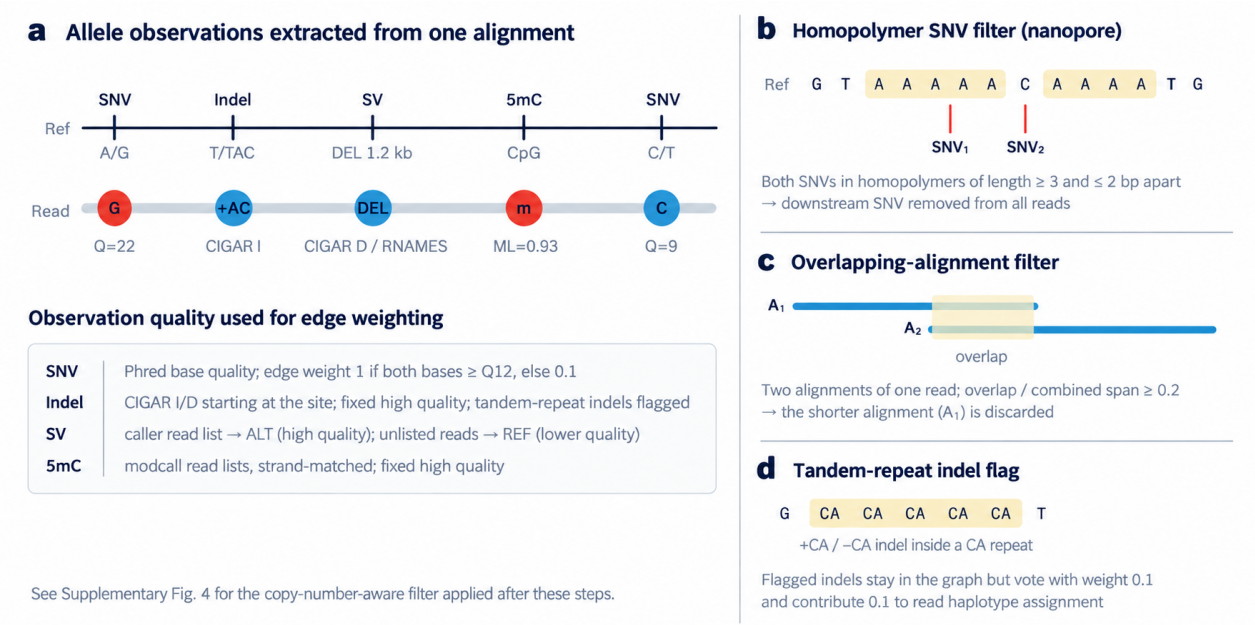
Supplementary Table 7: Software and data used in the benchmarks. [Complete versions and accessions.]

Item	Version	Notes
HG002 nanopore R10.4.1 reads	[–]	[flow cell, kit, basecaller model, source]
Reference	GRCh38 [build]	
minimap2	[–]	-ax map-ont -y
Clair3	[–]	SNV and indel calls
Sniffles2	[–]	-output-rnames
GIAB v4.2.1 HG002	v4.2.1	GRCh38-based small-variant benchmark
HG002 Q100 (T2T-based)	[v?]	referred to as v5.0q
WhatsHap	2.8	[options]
HapCUT2	1.3.4	[options]
LongPhase 1.0	1.0.x [exact]	-ont
LongPhase 2	[release]	-ont; gnn with defaults

Supplementary Table 8: Selected published tools, pipelines and studies that use LongPhase.
 Compiled from Europe PMC (80 indexed citing articles) and OpenAlex (96 citing works) for 10.1093/bioinformatics/btac058, September 2026; the “use” column quotes or paraphrases the citing article’s Methods.

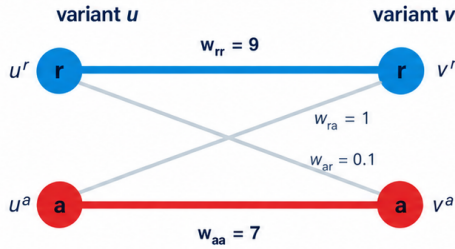
Work	Journal, year	Use of LongPhase
<i>Variant callers and pipelines</i>		
ClairS ¹	Nat. Methods 2026	germline phasing and haplotagging of tumour and normal alignments
ClairS-TO ²	Nat. Commun. 2025	phasing of input alignments
Clair3-RNA ³	Nat. Commun. 2025	phasing in the long-read RNA pipeline
Clair3 (accelerated) ⁴	Bioinformatics 2026	intermediate variant phasing and haplotagging
DeepVariant local haplotagging ⁵	Nat. Commun. 2024	cited as haplotagging method used by Clair3/Medaka
Severus ⁶	Nat. Biotechnol. 2025	haplotagged BAMs for somatic SV calling
EPI2ME wf-human-variation ⁷	ONT workflow	<code>-use_longphase</code> phasing
HiFi-somatic-WDL ⁸	PacBio workflow	optional phasing tool
cuteHap ⁹	Adv. Sci. 2026	read phasing before haplotype-aware SV calling
SpecImmune ¹⁰	Adv. Sci. 2026	joint SNV–SV phasing of immune genes
<i>Biological and clinical studies</i>		
Fluctuating DNA methylation in cancer ¹¹	Nature 2025	read phasing (v1.7) and <code>modcall</code> for CpG methylation
Advanced cancer cohort (POG) ¹²	Cell Genomics 2024	discussed as phasing method for tumours
Pediatric epilepsy epesignatures ¹³	Nat. Commun. 2024	phasing of Clair3 and SV calls
GNAS transcripts in PHP1B ¹⁴	PNAS 2025	phasing of Clair3 calls
3-hour genome for neonatal risk ¹⁵	Genet. Med. Open 2024	real-time phasing during sequencing
Ultrarapid diagnostics in critical care ¹⁶	Eur. J. Hum. Genet. 2025	read phasing (v1.0.7)
HYDIN in primary ciliary dyskinesia ¹⁷	Eur. J. Hum. Genet. 2024	phasing of Clair3 calls
Antisense oligonucleotides in APBD ¹⁸	Nucleic Acids Res. 2025	phasing of Clair3 calls
Native CpG methylation fine-mapping ¹⁹	HGG Adv. 2025	phasing and haplotagging (chosen for speed and SV support)
Canadian national platform benchmark ²⁰	Front. Genet. 2026	phasing inside the Clair3 workflow
Pallas’s cat genome ²¹	NAR Genom. Bioinform. 2023	haplotagging for allele-specific methylation
<i>Methods evaluated against LongPhase</i>		
HiPhase ²²	Bioinformatics 2024	comparison method
Ralphi ²³	Genome Res. 2025	comparison method (fastest runtime on ONT/PacBio)
GCphase ²⁴	BMC Bioinformatics 2024	comparison method
Duet ²⁵	BMC Bioinformatics 2022	comparison method for SV phasing

Supplementary Figures



Supplementary Fig. 1: Read-level allele observations and pre-graph filters. **a**, For every alignment, each overlapped variant yields an allele observation with an associated quality: SNVs from the aligned base (Phred quality), indels from CIGAR operations starting at the site, SVs from CIGAR operations of matching length or from the caller’s read list, and 5mC from the `modcall` read lists. The table summarizes how each class enters the edge weights. **b**, Nanopore-specific homopolymer filter: when two SNVs both lie in homopolymers of length ≥ 3 and are ≤ 2 bp apart, the downstream SNV is removed. **c**, Overlapping-alignment filter for supplementary alignments of one read. **d**, Indels in tandem repeats are flagged and vote with weight 0.1.

a Four allele-pair weights between two variants



$$P = w_{rr} + w_{aa} = 16 \text{ (cis: same haplotype carriers } r \text{ and } r)$$

$$Q = w_{ra} + w_{ar} = 1.1 \text{ (trans)}$$

$$\text{similarity } s = \min(P, Q) / \max(P, Q) = 0.07$$

Per read: +1 if both bases $\geq Q12$, else +0.1 (grey edges here carry a low-quality read).

b Vote rules for the pair (u, v)

Condition on s and variant type	Vote rule
$s > 0.7$	no vote (uninformative pair)
$s > 0.3$ for SNV-5mC pairs	no vote
$0.1 < s \leq 0.7$	vote for larger of P, Q; weight 1
$s \leq 0.1$, or one side has no support	vote; weight 20 (near-unanimous)
u is a tandem-repeat indel	vote; weight 0.1

The vote is expressed as a haplotype for v using the haplotype already assigned to u: P favours v^r on $\text{hap}(u^r)$, Q favours v^r on $\text{hap}(u^a)$.

c Single-read guard

Among the votes received by v, count those whose pair is supported by a single read ($P + Q \leq 1$). If more than three such votes exist, h_1 and h_2 are recomputed from high-consistency ($s < 0.2$), non-indel votes only, preventing one chimeric ultra-long read from dominating a sparse region.

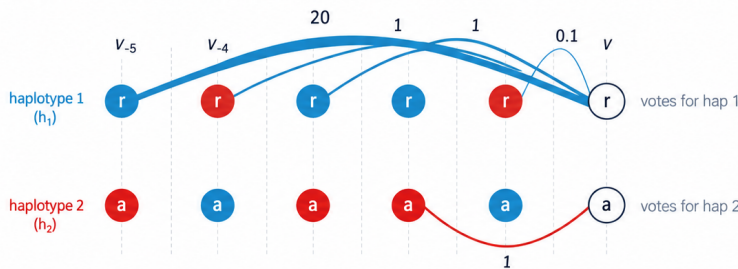
d Exported graph (DOT)

```
1001.1 -> 1543.1 [label=16.0]
1001.2 -> 1543.2 [label=1.1]
```

position.allele → position.allele, label = accumulated haplotype vote weight at the source;
one edge per voting pair; parsed by the GNN module together with INFO/PE, H1, H2.

Supplementary Fig. 2: Pairwise allele support and vote weighting. **a**, Between two variants u and v the reads accumulate four weights ($w_{rr}, w_{ra}, w_{ar}, w_{aa}$); each read adds 1 when both bases have quality $\geq Q12$ and 0.1 otherwise. P and Q are the supports for the two possible pairings and s their similarity. **b**, Decision rules that turn (P, Q, s) into a weighted haplotype vote for v . **c**, Guard against a single erroneous ultra-long read. **d**, Format of the exported DOT graph consumed by the **gnn** module.

a Variant v receives weighted votes from up to k = 35 upstream variants



$$h_1(v) = 20 + 1 + 1 + 0.1 = 22.1 \quad h_2(v) = 1$$

v assigned to haplotype 1; tie $h_1 = h_2$ opens a new phase block (PS = position of first variant)

b Phasing entropy (PE)

$$PE(v) = -\sum p_i \log_2 p_i, \quad p_i = h_i / (h_1 + h_2)$$

$$22.1 : 1 \quad PE = 0.257$$

$$3 : 1 \quad PE = 0.811$$

$$4 : 3 \quad PE = 0.985$$

$$5 : 5 \quad PE = 1.000$$

PE ≥ 0.80 triggers a GNN window (Supplementary Fig. 6).

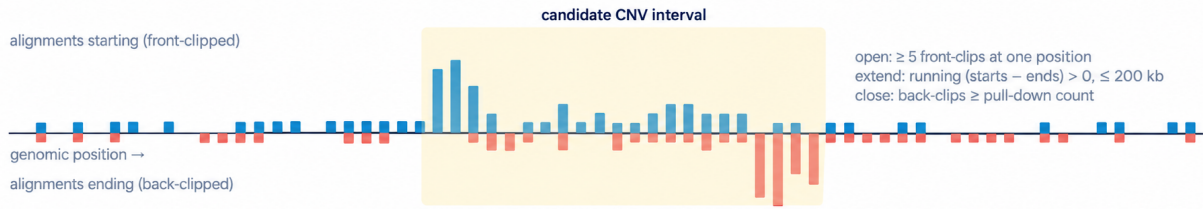
Written to VCF as INFO/PE, INFO/H1, INFO/H2 for every phased variant.

c Block formation

Variants are visited in coordinate order. Each assigned variant casts votes to its next k neighbours within 300 kb; a variant that receives no informative vote, or a tie, starts a new block. Blocks with a single variant are dropped. Haplotype labels are propagated along the block to produce phased GTs (0/1 | 1/0).

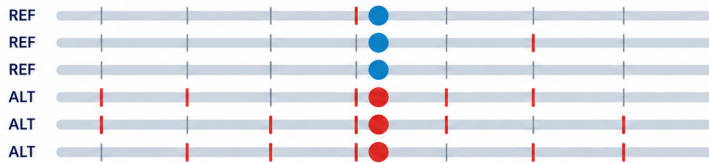
Supplementary Fig. 3: Multi-neighbour voting and phasing entropy. **a**, Variant v receives weighted votes from up to $k = 35$ upstream variants within 300 kb; line thickness encodes vote weight (20, 1 or 0.1). The haplotype with the larger total wins; a tie opens a new phase block. **b**, Phasing entropy of the vote totals for four examples; values ≥ 0.80 (dashed box) trigger a GNN window. **c**, Block formation and propagation of haplotype labels.

a Clipping-based detection of copy-number-altered intervals



b Removing heterozygous calls whose ALT reads carry the mismatches

Variant inside interval; other heterozygous sites on the same reads shown as ticks



mean mismatches per read

REF-supporting reads: 0.7

ALT-supporting reads: 4.7

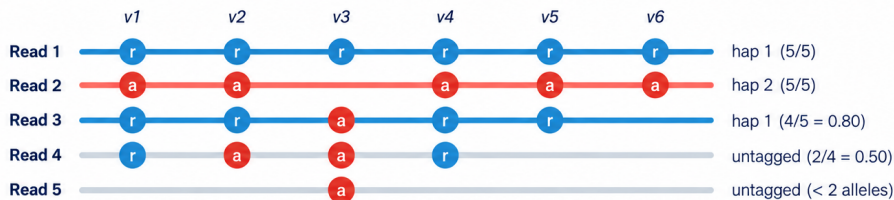
ratio = ALT / (REF + ALT) = 0.87 ≥ 0.7

→ site removed from all reads before graph construction

Targets heterozygous calls created by collapsed paralogs or copy-number change, whose ALT allele co-occurs with a paralogue-specific mismatch pattern.

Supplementary Fig. 4: Copy-number-aware filtering of unreliable heterozygous sites. a, Per-position counts of alignments starting (front-clipped, above the axis) and ending (back-clipped, below) are scanned by a finite-state detector that opens a candidate interval at ≥ 5 front-clips, extends it while starts exceed ends, and closes it when a comparable number of back-clips is observed (maximum 200 kb). b, Inside an interval, each heterozygous site is scored by the mean number of mismatches to the phasing graph on ALT- versus REF-supporting reads; sites whose ALT reads carry $\geq 70\%$ of the mismatches are removed from all reads, because their alternate allele is a paralogue-specific difference rather than a heterozygous variant.

a Step 1: assign each read to a haplotype from the initial phase



Read tagging rule: A read is tagged when $\max(n_1, n_2) / (n_1 + n_2) > 0.65$ and it carries ≥ 2 informative alleles; SNV and SV alleles count 1, indels 0.1, 5mC 0.

n_1, n_2 : counts of alleles supporting haplotype 1 and haplotype 2 (informative alleles only).

b Step 2: re-derive each variant's phase from tagged reads

Variant v3 (example of insufficient confidence)

hap-1 reads with REF + hap-2 reads with ALT = $n_1 = 1$

hap-1 reads with ALT + hap-2 reads with REF = $n_2 = 2$

$$\text{confidence } p = \frac{\max(n_1, n_2)}{(n_1 + n_2)} = \frac{2}{3} = 0.67 \leq 0.75$$

Conclusion: v3 left unphased (GT 0/1, no PS)

Variant v1 (example of confident assignment)

$n_1 = 3, n_2 = 0, p = 1.00 > 0.75$

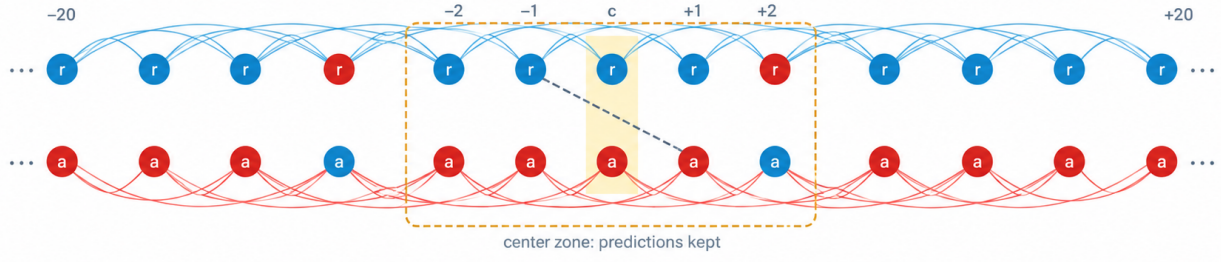
$$\text{confidence } p = \frac{\max(n_1, n_2)}{(n_1 + n_2)} = \frac{3}{3} = 1.00 > 0.75$$

→ phase confirmed: REF on hap 1, ALT on hap 2 (GT 0/1)

Only variants supported by the two read populations keep a phase; assignments driven by a few chimeric or mis-mapped reads are removed. Thresholds: --readConfidence 0.65, --snpConfidence 0.75.

Supplementary Fig. 5: Read-based correction. a, Each read is assigned to a haplotype by majority over its phased SNV and SV alleles (indels count 0.1, 5mC 0); reads with majority ≤ 0.65 or fewer than two informative alleles remain untagged. b, Each variant's phase is re-derived from the tagged reads and retained only if the consistent fraction exceeds 0.75; otherwise the variant is unphased.

a A window is opened around every phased variant with $PE \geq 0.8$



- up to 20 variants on each side (≤ 41 variants, ≤ 82 allele nodes; windows above 256 nodes are skipped)
- nodes: one per allele; SNV, indel, SV and 5mC records are merged and sorted by position
- edges: DOT edges with both endpoints in the window, symmetrized, plus one self-loop per node (mean of incident edge features)
- predictions of the two allele nodes are averaged; a variant covered by several windows receives the mean

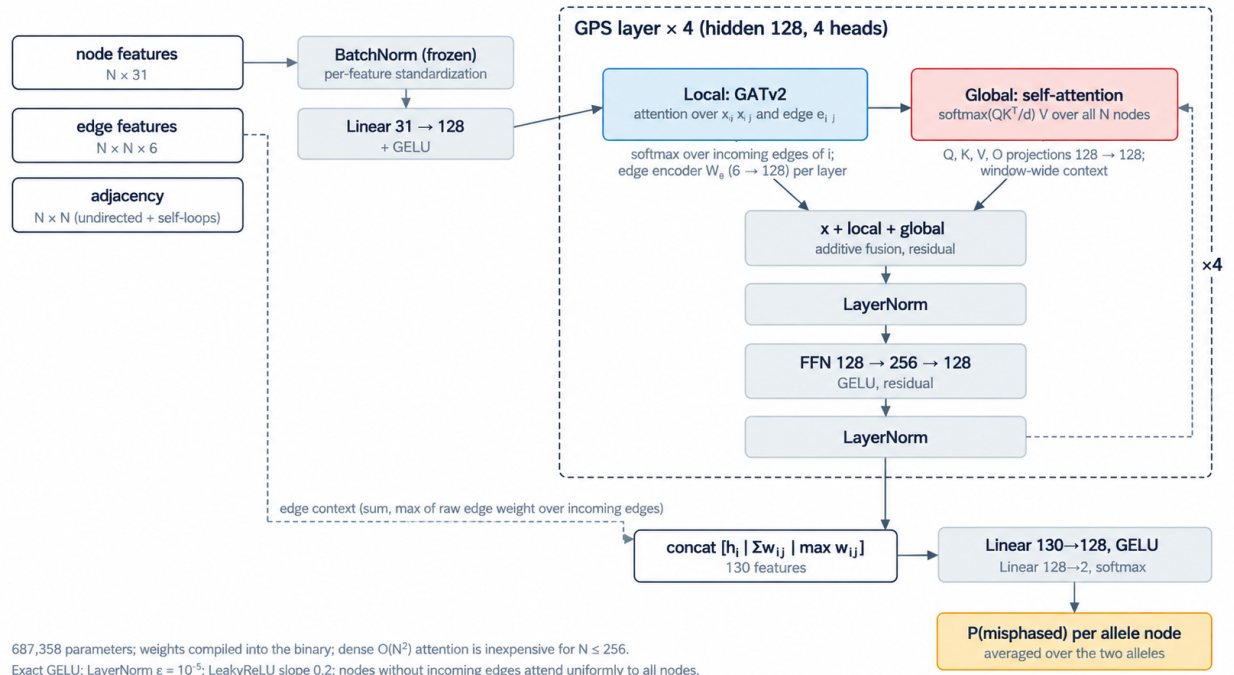
b Bridge vertex feature



c Center-zone mask

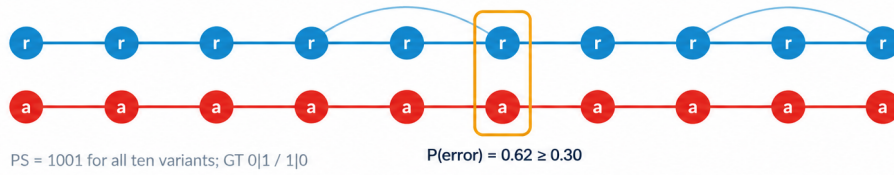
$r = |\text{pos} - \text{pos}_c| / \text{max offset in window}$
kept if $r \leq \min(1, 10 / \text{round}(20 \cdot r_{\text{max}}))$
so that predictions are read only where both sides of the variant are visible to the network.

Supplementary Fig. 6: Window construction for the graph neural network. **a**, A window of ± 20 variants (two allele nodes each) is opened around every phased variant with $PE \geq 0.8$; edges are the DOT edges with both endpoints in the window (blue and red arcs) plus self-loops; a dashed grey edge marks a cross-haplotype link. Predictions are read only in the central zone. **b**, Bridge-vertex feature: a variant whose removal disconnects the window graph. **c**, Definition of the central zone.

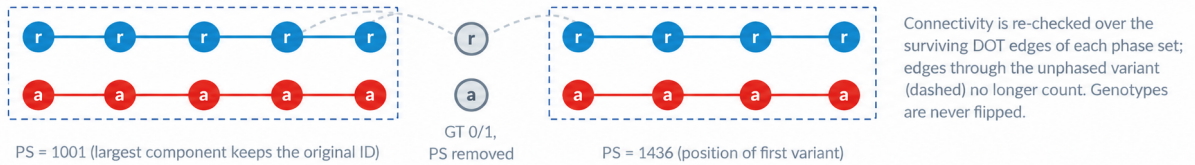


Supplementary Fig. 7: Graph neural network architecture. Node features are standardized and projected to 128 dimensions. Each of four GPS layers combines a local GATv2 branch over the window graph (edge features enter through a per-layer edge encoder) with a global self-attention branch over all nodes, fused additively into the residual stream and followed by layer normalization and a feed-forward block. The final embedding is concatenated with the sum and maximum of incoming raw edge weights and classified into phased-correctly versus misphased; the probabilities of the two allele nodes of a variant are averaged.

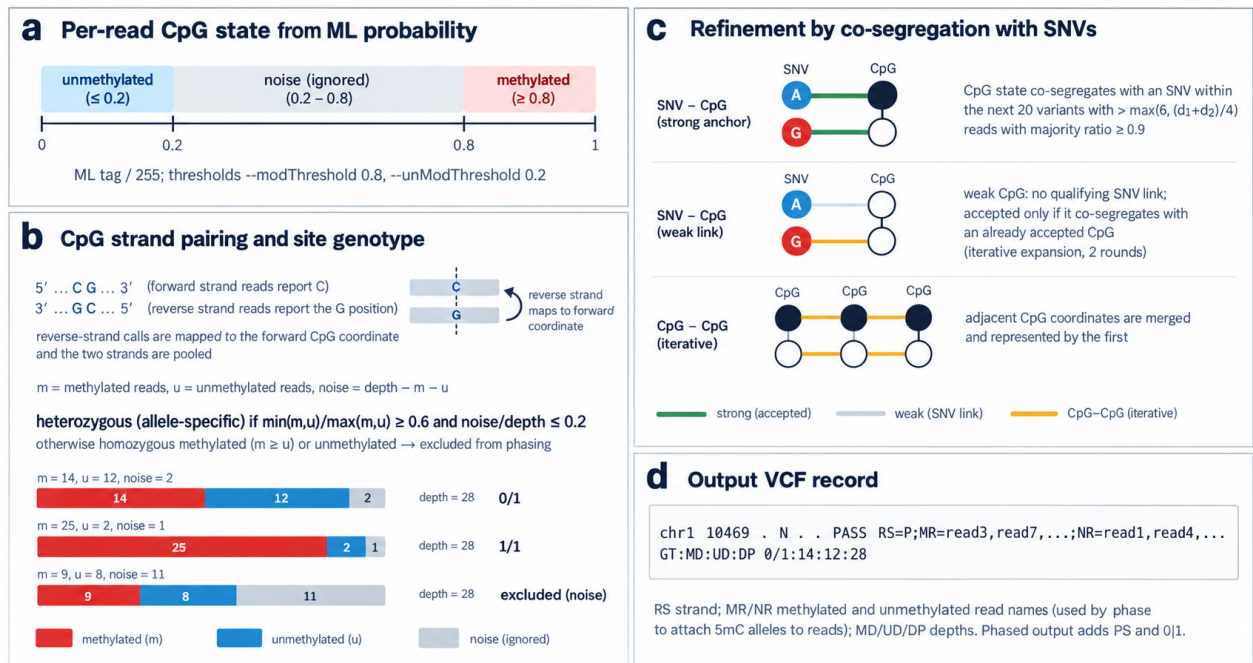
a Before correction: one phase set



b After correction: variant unphased, block split



Supplementary Fig. 8: Unphasing and phase-set splitting. **a**, A variant with predicted error probability ≥ 0.30 (amber) inside a phase set. **b**, After correction the variant is written as an unphased genotype without PS; connectivity of the remaining variants is recomputed over the surviving DOT edges, the largest component keeps the original PS and every other component receives a new PS equal to the position of its first variant. Genotypes are never flipped.



Supplementary Fig. 9: Allele-specific methylation calling with modcall. **a**, Per-read CpG states from the ML probability (methylated ≥ 0.8 , unmethylated ≤ 0.2 , otherwise noise). **b**, Reverse-strand calls are mapped to the forward CpG coordinate, strands are pooled, and a site is heterozygous when the minority state is at least 60% of the majority state and noise is at most 20% of depth. **c**, Candidate heterozygous CpGs are refined by co-segregation with SNVs on the same reads (strong anchors) or with already accepted CpGs (weak, iterative). **d**, Output record with methylated and unmethylated read names used by phase.

a Switch errors, flips and Hamming distance within one block

variant position	1	2	3	4	5	6	7	8	9	10
truth h_0	0	0	0	0	0	0	0	0	0	0
query h_0	0	0	0	1	1	1	1	0	1	0
switch encoding (query)		0	0	1	0	0	0	1	1	1

one flip (positions 7–8)

Switch errors
= number of positions where the switch encodings differ
= 4

Switch/flip decomposition
two adjacent switches (positions 7–8) form one flip; isolated switches = 2, flips = 1

Hamming distance (orientation-aware)
= min over the two orientations of misassigned variants
= $\min(5, 5) = 5$ of 10 (50%)

Block-wise aggregation
reported per-block minimum, summed over blocks, divided by variants compared

b Block statistics

Intersected blocks

Blocks are intersected: only variants heterozygous in both files and phased in both are compared; singleton blocks are ignored.

Phased fraction (SNV %)

= truth heterozygous SNVs phased by the query / all truth heterozygous SNVs (same for indels, SVs, 5mC).

Block N50

= span of the block at which the cumulative span of query blocks, sorted by length, reaches 50% of the total.

Switch error rate (%)

= switch errors / assessed variant pairs $\times 100$.
Cross-tool comparisons use SNVs only ($\sim$ only-snvs).

Implementation note: Implemented in `longphase compare` (re-implementation of `whatshap compare`; identical values, parallel over chromosomes).

Supplementary Fig. 10: Phasing evaluation metrics. **a**, Within one intersected block, the query haplotype string is compared with the truth in switch encoding; the number of differing positions is the switch-error count, adjacent switch pairs are counted as flips, and the Hamming distance is the minimum over the two orientations of misassigned variants. **b**, Definitions of block intersection, phased fraction, block N50 and switch error rate as implemented in `longphase compare`.

References

- [1] Zheng, Z. *et al.* ClairS: a deep-learning method for long-read tumor–normal pair somatic small variant calling. *Nature Methods* **23**, 1507–1520 (2026).
- [2] Chen, L. *et al.* ClairS-TO: a deep-learning method for long-read tumor-only somatic small variant calling. *Nature Communications* **16** (2025).
- [3] Zheng, Z. *et al.* Clair3-RNA: a deep learning-based small variant caller for long-read RNA sequencing data. *Nature Communications* **16** (2025).
- [4] Zheng, Z. *et al.* Accelerated long-read variant calling with Clair3 for whole-genome sequencing. *Bioinformatics* **42** (2026).
- [5] Kolesnikov, A. *et al.* Local read haplotagging enables accurate long-read small variant calling. *Nature Communications* **15** (2024).
- [6] Keskus, A. G. *et al.* Severus detects somatic structural variation and complex rearrangements in cancer genomes using long-read sequencing. *Nature Biotechnology* **44**, 247–257 (2025).
- [7] Oxford Nanopore Technologies. epi2me-labs/wf-human-variation: Nextflow workflow for the analysis of human genetic variation from Oxford Nanopore sequencing data. <https://github.com/epi2me-labs/wf-human-variation> (2025). Version 2.x; phasing performed with LongPhase (`-use_longphase`).
- [8] Pacific Biosciences. HiFi-somatic-WDL: tumor-normal variant calling workflow using HiFi reads. <https://github.com/PacificBiosciences/HiFi-Somatic-WDL> (2025).
- [9] Cao, S. *et al.* cutehap: Haplotype-Aware structural variant detection in phased Long-Read sequencing data. *Advanced Science* **13** (2026).
- [10] Wang, S. *et al.* A scalable framework for comprehensive typing of polymorphic immune genes from Long-Read data. *Advanced Science* **13** (2026).
- [11] Gabbutt, C. *et al.* Fluctuating DNA methylation tracks cancer evolution at clinical scale. *Nature* **645**, 764–773 (2025).
- [12] O’Neill, K. *et al.* Long-read sequencing of an advanced cancer cohort resolves rearrangements, unravels haplotypes, and reveals methylation landscapes. *Cell Genomics* **4**, 100674 (2024).
- [13] LaFlamme, C. W. *et al.* Diagnostic utility of DNA methylation analysis in genetically unsolved pediatric epilepsies and CHD2 epismature refinement. *Nature Communications* **15** (2024).
- [14] Iwasaki, Y. *et al.* Bidirectional disruption of *GNAS* transcripts causes broad methylation defects in pseudohypoparathyroidism type 1b. *Proceedings of the National Academy of Sciences* **122** (2025).
- [15] Zalusky, M. P. *et al.* 3-hour genome sequencing and targeted analysis to rapidly assess genetic risk. *Genetics in Medicine Open* **2**, 101833 (2024).
- [16] Smits, D. J. *et al.* Nanopore long-read sequencing for the critically ill facilitates ultrarapid diagnostics and urgent clinical decision making. *European Journal of Human Genetics* **34**, 108–118 (2025).
- [17] Fleming, A. *et al.* Combined approaches, including long-read sequencing, address the diagnostic challenge of HYDIN in primary ciliary dyskinesia. *European Journal of Human Genetics* **32**, 1074–1085 (2024).

- [18] Thomas, R. *et al.* Splice-modulating antisense oligonucleotides targeting a pathogenic intronic variant in adult polyglucosan body disease correct mis-splicing and restore enzyme activity in patient cells. *Nucleic Acids Research* **53** (2025).
- [19] Tian, Y., McDonnell, S. K., Wu, L., Larson, N. B. & Wang, L. Fine mapping regulatory variants by characterizing native CpG methylation with nanopore long-read sequencing. *Human Genetics and Genomics Advances* **7**, 100532 (2026).
- [20] Galvez, J. H. *et al.* Benchmarking of Oxford Nanopore whole genome sequencing for germline variant and CpG methylation detection across Canada’s national platform for genome sequencing and analysis. *Frontiers in Genetics* **17** (2026).
- [21] Flack, N. *et al.* Chromosome-level, nanopore-only genome and allele-specific DNA methylation of Pallas’s cat, *Otocolobus manul*. *NAR Genomics and Bioinformatics* **5** (2023).
- [22] Holt, J. M. *et al.* HiPhase: jointly phasing small, structural, and tandem repeat variants from HiFi sequencing. *Bioinformatics* **40** (2024).
- [23] Battistella, E., Maheshwari, A., Ekim, B., Berger, B. & Popic, V. Graph-based deep reinforcement learning for haplotype assembly with Ralphi. *Genome Research* **35**, 2617–2625 (2025).
- [24] Luo, J., Wang, J., Zhai, H. & Wang, J. GCphase: an SNP phasing method using a graph partition and error correction algorithm. *BMC Bioinformatics* **25** (2024).
- [25] Zhou, Y., Leung, A. W.-S., Ahmed, S. S., Lam, T.-W. & Luo, R. Duet: SNP-assisted structural variant calling and phasing using Oxford nanopore sequencing. *BMC Bioinformatics* **23** (2022).